Random Forest

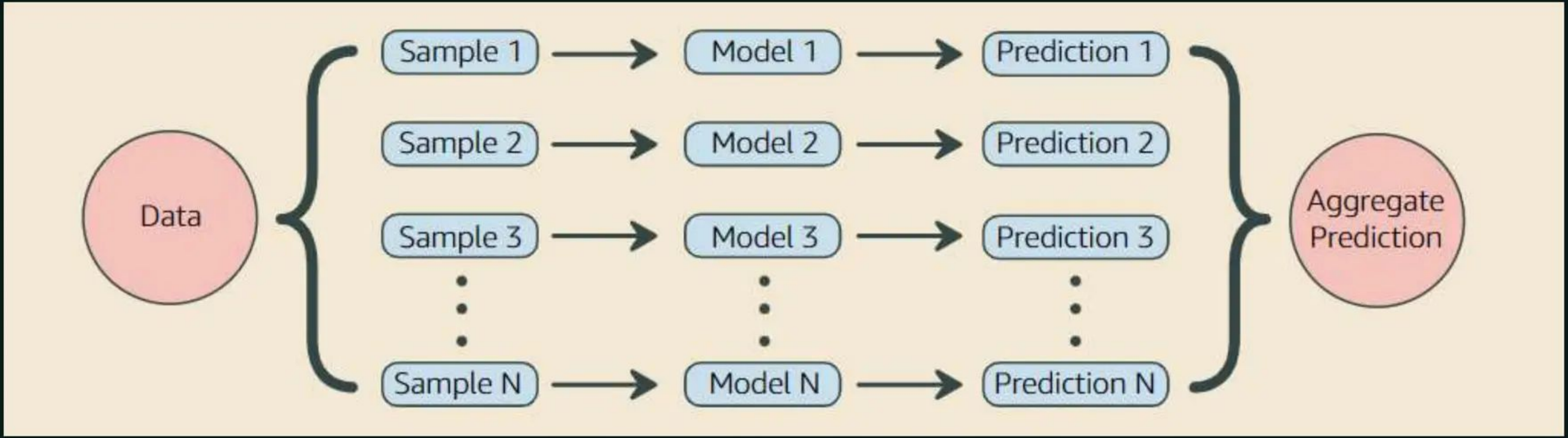
Ensemble methods

Ensemble learning combines multiple models to improve predictive performance.

- **Types of Ensemble Methods**:
  - **Bagging** (Bootstrap Aggregating) → Random Forest
  - **Boosting** (e.g., AdaBoost, Gradient Boosting)
  - **Stacking**
- **Why Ensemble Learning?**
  - Reduces overfitting
  - Improves accuracy
  - Handles bias-variance trade-off

Ensemble Learning

Ensemble learning creates a stronger model by aggregating the predictions of multiple weak models. Random Forest is an example of ensemble learning where each model is a decision tree. The idea behind it is- the wisdom of the crowd. The majority vote aggregation can have better accuracy than the individual models.



Dataset Preparation

Bootstrap Aggregating (Bagging)

Creating a different training subset randomly from the original training dataset with replacement is called Bagging. With replacement refers to the bootstrap sample having duplicate elements. Reduces variance, helps to avoid overfitting.

Out Of Bag (OOB)

The out-of-bag dataset represents the remaining elements which were not in the bootstrap dataset. Used for cross validation or to evaluate the performance.

Random Subspace Method(Feature Bagging)

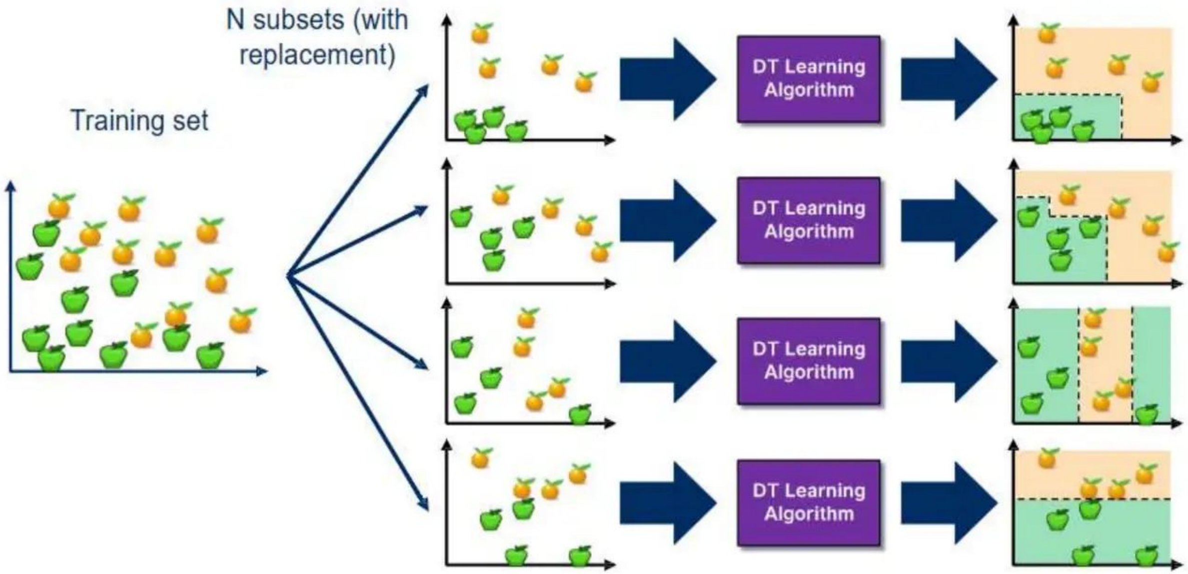
While building the tree, for splitting, a randomly selected subset of the features are used. So, the trees are more different having low correlation.

What Is Random Forest?

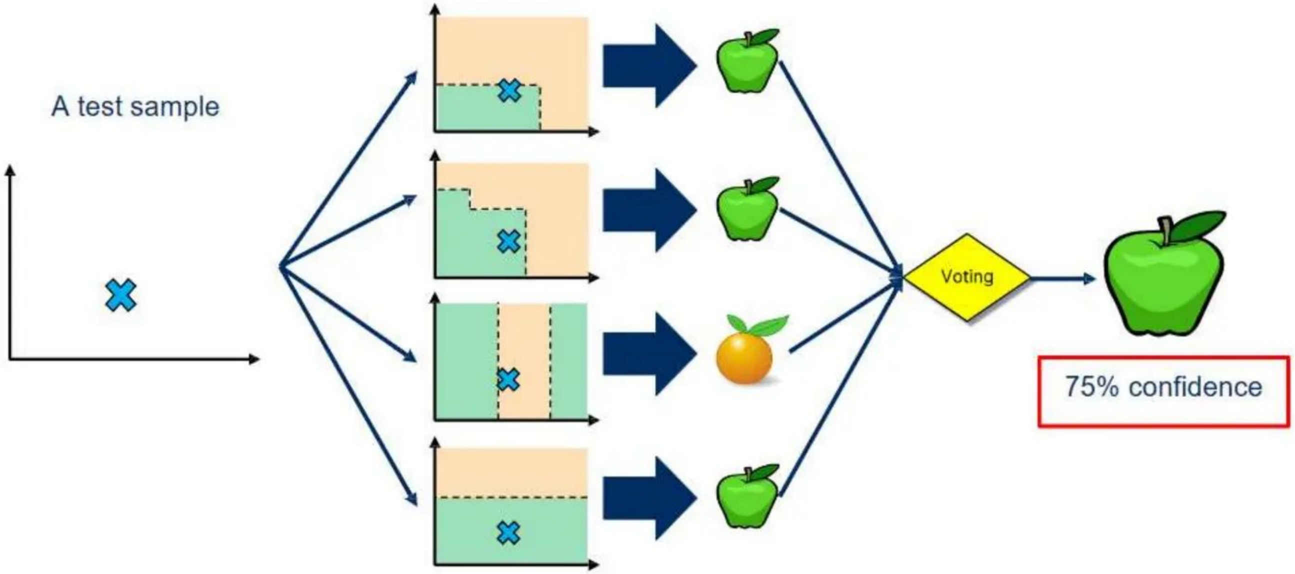
Random forest is a supervised learning algorithm. It is a machine learning algorithm that creates an ensemble of multiple decision trees to reach a singular, more accurate prediction or result.

- **Key Features**:
  - Uses **bagging** (bootstrap samples)
  - Introduces **random feature selection** for decorrelation
  - Outputs the **majority vote** (classification) or **average** (regression)

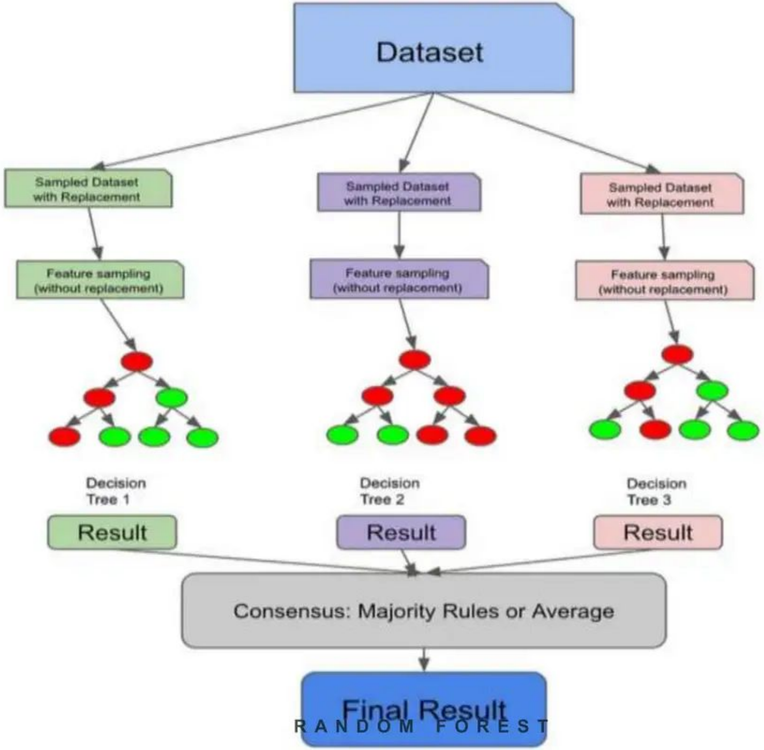
Bagging at training time



Bagging at inference time



Random Forest Model



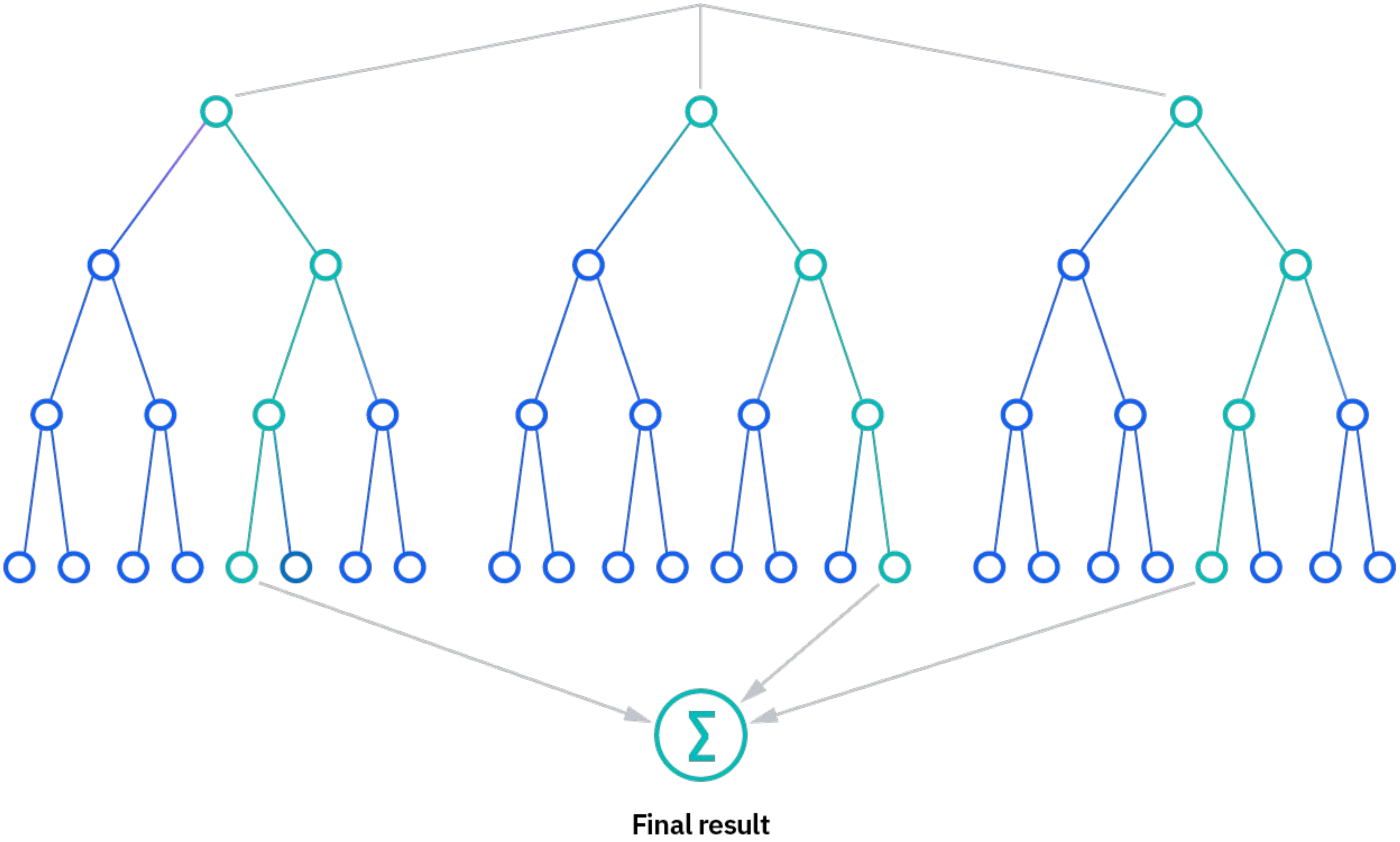
Random Forest Applications

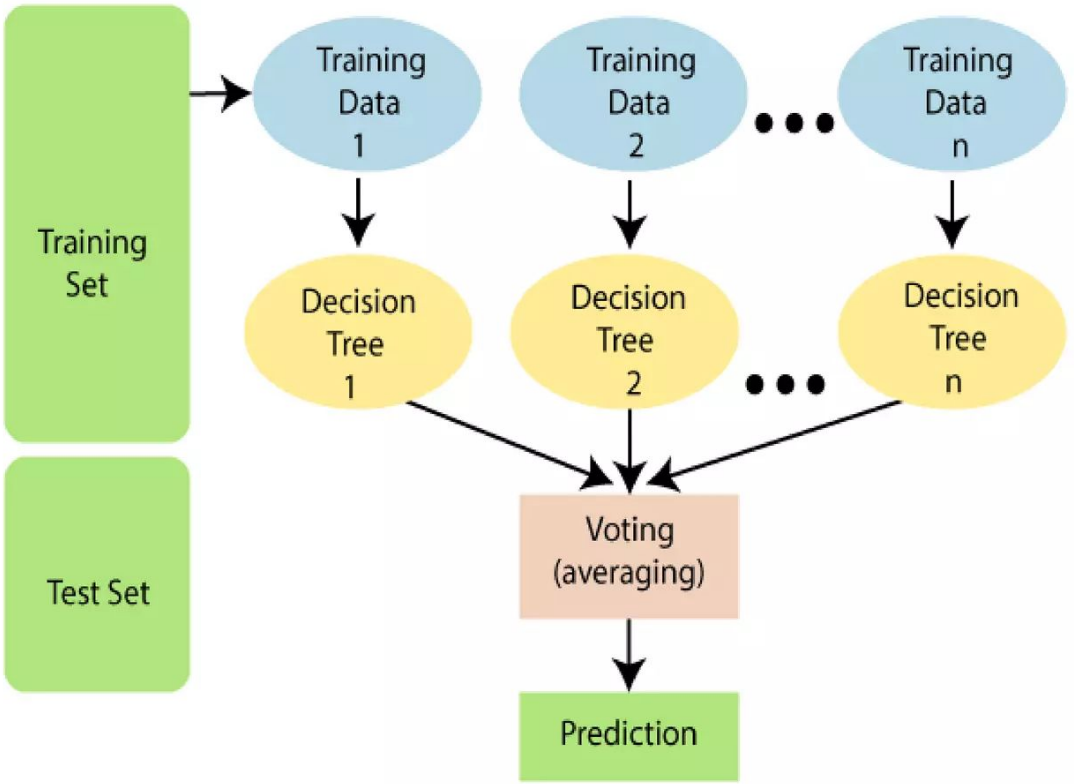
- Detects reliable debtors and potential fraudsters in finance
- Verifies medicine components and patient data in healthcare
- Gauges whether customers will like products in e-commerce

How Random Forest Works

Random forest algorithms have three main hyperparameters, which need to be set before training. These include node size, the number of trees, and the number of features sampled. From there, the random forest classifier can be used to solve for regression or classification problems.

The random forest algorithm is made up of a collection of decision trees, and each tree in the ensemble is comprised of a data sample drawn from a training set with replacement, called the bootstrap sample. Of that training sample, one-third of it is set aside as test data, known as the out-of-bag (oob) sample. Another instance of randomness is then injected through feature bagging, adding more diversity to the dataset and reducing the correlation among decision trees. Depending on the type of problem, the determination of the prediction will vary. For a regression task, the individual decision trees will be averaged, and for a classification task, a majority vote—i.e. the most frequent categorical variable—will yield the predicted class. Finally, the oob sample is then used for cross-validation, finalizing that prediction.


Σ
Final result


Training Set
Test Set
Training Data 1
Training Data 2
Training Data n
Decision Tree 1
Decision Tree 2
Decision Tree n
Voting (averaging)
Prediction

Aspect	Decision Tree	Random Forest
Overfitting	High risk	Reduced via bagging
Accuracy	Lower	Higher
Speed	Faster	Slower (but parallel)
Interpretability	High	Low